Complementary Supervised and Benign-Only Anomaly Detection for Unseen SQL Injection Families at the Database Proxy

1st Minh-Duc Do-Xuan
International School
Vietnam National University Hanoi
Hanoi, Vietnam
minhdc42195@gmail.com

2nd Bach Luong-Chi
School of Science, Engineering & Technology
RMIT University Vietnam
Hanoi, Vietnam
chibachluong@gmail.com

3rd Diep Dinh-Ngoc
International School
Vietnam National University Hanoi
Hanoi, Vietnam
dinhngocdiep281104.work@gmail.com

4th Minh Nguyen-Quang
International School
Vietnam National University Hanoi
Hanoi, Vietnam
quangminhdang1703@gmail.com

5th Linh Dinh-Van
Academy of Cryptography Techniques
Hanoi, Vietnam
vanlinh@actvn.edu.vn

6th Thai Dinh Kim*
International School
Vietnam National University Hanoi
Hanoi, Vietnam
thaikd@vnu.edu.vn

***Abstract*—SQL Injection (SQLi) remains a persistent web-application threat, while detectors trained on fixed attack labels can generalize unevenly to attack families absent from supervised training. This paper studies a two-branch SQLi detector intended for deployment at the application–database boundary. Branch 1 performs supervised multi-class classification, whereas Branch 2 learns only from benign traffic and scores query-level structural anomalies. Across a six-class benchmark assembled from four public data sources, the selected DistilBERT classifier reaches a macro-F1 of 0.9699 on the standard held-out test split. Local Outlier Factor (LOF) achieves an area under the curve (AUC) of 0.915 and detects 78.42% of held-out anomalous queries at a 5.27% false-positive rate (FPR). Generalization is further examined with leave-one-attack-family-out evaluation for union-based, error-based, boolean-blind, and time-blind SQLi, with both branches scoring the same withheld attack queries. Union-based, error-based, and time-blind attacks retain 99.8–100% combined coverage. In contrast, Branch 1 maps 86.5% of unseen boolean-blind queries to normal; benign-only anomaly scoring recovers many of these cases and raises combined coverage to 88.8%, leaving an 11.2% residual gap. These results show that closed-set classification accuracy can mask attack-specific generalization failures and quantify when benign-only anomaly detection contributes complementary coverage.**

***Index Terms*—SQL injection, unseen attack families, anomaly detection, DistilBERT, Local Outlier Factor, database proxy, cybersecurity**

I. Introduction

Web applications store increasingly sensitive data, making their back-end databases a persistent target. SQL Injection (SQLi), in which unsanitized user input changes the semantics of a database query, remains one of the oldest and most damaging web vulnerabilities and is represented in the OWASP injection category [1]. Successful exploitation can disclose, alter, or destroy data and may enable privilege escalation. Web Application Firewalls (WAFs) and signature-based rules are effective for known patterns, but obfuscation and previously unseen payloads can reduce their coverage, while overly strict rules can raise false positives on legitimate traffic [2], [3].

Machine learning (ML) and deep learning provide a complementary direction because they learn statistical and semantic patterns from queries or payloads rather than depending only on hand-written signatures [2], [4], [5]. Classical approaches use TF-IDF, character or word n-grams, and structural features, while deep models use CNN, recurrent, and Transformer encoders [5]–[9]. Recent SQLi studies have reported high closed-set performance with BERT-family models and lightweight attention architectures [9]. These results establish strong recognition of attack categories represented in training, but they do not directly measure what happens when an entire SQLi family is absent from the supervised label space.

This paper focuses on class-level generalization when an attack family is absent from supervised training. The proposed system combines two query-level signals at the application–database boundary: supervised multi-class classification for learned SQLi patterns and benign-only anomaly detection for deviations from normal query structure. Beyond the standard closed-set benchmark, four attack families are evaluated under leave-one-attack-family-out training. Both branches score the same withheld queries, allowing the analysis to distinguish supervised transfer, anomaly-based recovery, and cases missed by both mechanisms.

The contributions of this paper are:

- a two-branch SQLi detection pipeline that combines supervised six-class classification with benign-only anomaly scoring over a heterogeneous four-source benchmark;

*Corresponding author: Thai Dinh Kim (thaikd@vnu.edu.vn).

- a campaign-aware split for the large `sqlmap`-derived source to reduce near-duplicate leakage, together with a benign pool spanning structurally different query distributions;
- a paired leave-one-attack-family-out evaluation over four SQLi families that reports supervised miss rate, benign-only anomaly detection rate, and their combined coverage on identical withheld queries; and
- an attack-specific generalization analysis showing a pronounced gap for boolean-blind SQLi despite its near-perfect closed-set classification performance.

The remainder of the paper reviews related work in Section II, describes the two-branch detection framework in Section III, presents the datasets and experimental setup in Section IV, reports the experimental results in Section V, discusses the findings and limitations in Section VI, and concludes the paper in Section VII.

II. Related Work

Traditional SQLi defenses include parameterized queries, input validation, static signatures, and WAF rule sets such as the OWASP Core Rule Set [3]. Learning-based alternatives have used SVM, Random Forest, XGBoost, Logistic Regression, feature selection, and ensemble methods [4], [10], [11]. Deep models further learn query representations directly: CNNs capture local token patterns [5], contextualized embeddings improve SQLi classification [7], [8], and lightweight self-attention models have been evaluated against ALBERT, DistilBERT, and ELECTRA on SQLi queries [9].

Anomaly detection addresses a different setting by learning benign behavior and flagging deviations. One-Class SVM [12] and Isolation Forest [13] are standard one-class baselines, while recent work has trained deep anomaly ensembles exclusively on normal web requests for zero-day web-attack detection [14]. Superviz25-SQL was introduced specifically to support unsupervised SQLi evaluation and provides nine baseline detection pipelines over a large, realistic `sqlmap`-generated corpus [15]. Hybrid SQLi systems have also combined supervised sequence models with benign-trained anomaly components to improve coverage of known and previously unseen queries [16].

Adversarial studies such as WAF-A-MoLE show that learned and rule-based detectors can also be challenged by semantics-preserving transformations [17]. Taken together, prior work establishes supervised SQLi classification, benign-only anomaly detection, and hybrid detection as useful components. The present study examines their complementarity under attack-family exclusion. For four SQLi families, the supervised classifier is retrained without the target family and both branches are then evaluated on the same withheld queries. This paired design exposes supervised transfer, anomaly-only recovery, and joint failure at the query level.

Table I places the experiment within these evaluation settings. The paired class-exclusion protocol decomposes unseen-family performance into supervised transfer, anomaly detection on identical queries, and their union. The contribution of the benign-only branch can therefore be observed directly for each evaluated withheld family rather than only through an aggregate fused score.

TABLE I
Representative recent evaluation settings for learned SQLi/web-attack detection.

Study	Learning setting	Generalization evaluation
Lo et al. [9]	supervised attention; BERT-family baselines	closed-set SQLi plus long-query length shift
Babaey and Faragardi [14]	benign-only deep anomaly ensemble	web-attack families detected without attack training
Quetel et al. [15]	unsupervised/novelty SQLi baselines	attack-technique benchmarking on Superviz25-SQL
Okello et al. [16]	supervised CNN–LSTM + benign-trained autoencoder	hybrid known/unseen SQLi detection
This work	supervised multi-class + benign-only LOF	paired family exclusion; both branches score identical unseen queries

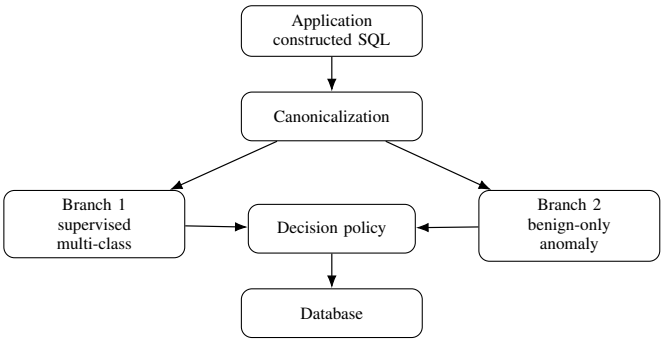


Fig. 1. Two-branch query-level detector and intended database-proxy placement.

III. Two-Branch Query-Level SQLi Detection

A. System Placement and Canonicalization

The intended deployment point is the application–database boundary, where a proxy can inspect a constructed SQL statement before execution. At this point, detection can operate on query-level content after application-side processing rather than relying solely on raw HTTP syntax. The experimental benchmark described in Section IV includes both SQL statements and HTTP-derived parameter content, which are canonicalized into a common query-level representation before feature extraction.

Each query is canonicalized before feature extraction. Common encodings (URL, hexadecimal, and `CHAR()`/`ASCII()` forms) are decoded, SQL keyword casing is normalized, and comment sequences are marked rather than deleted. The purpose is to collapse syntactically equivalent forms while retaining cues associated with evasion. Figure 1 summarizes the detection pipeline and its intended placement.

B. Branch 1: Supervised Multi-Class Classification

Branch 1 assigns each query to one of six classes: `normal`, `union_based`, `error_based`, `boolean_blind`, `time_blind`, and `stacked`. Four candidate architectures

TABLE II
QUERY-LEVEL DECISION POLICY.

Branch 1	Branch 2	Action
attack class	any	block and log
normal	anomalous	hold/review
normal	normal	allow

are compared: TF-IDF+Logistic Regression, TF-IDF+LightGBM, DistilBERT, and a compact CNN with an SQL-oriented tokenizer. Multi-class output is useful operationally because it distinguishes the detected attack technique, and experimentally because it exposes whether an unseen attack is mapped to another attack class or incorrectly absorbed into `normal`.

C. Branch 2: Benign-Only Anomaly Detection

Branch 2 is trained exclusively on benign traffic and emits a continuous anomaly score. Branch 2 uses lightweight query statistics including length, special-character ratio, SQL-keyword count, Shannon entropy, quote imbalance, longest same-type token run, maximum token length, token count, longest special-character run, longest digit run, and parenthesis imbalance.

One-Class SVM, Isolation Forest, and Local Outlier Factor (LOF) are compared. A global One-Class SVM boundary performs poorly when benign traffic spans structurally different sources, and an explored per-source ensemble produces an overall FPR of approximately 41%. LOF is selected because its local-density criterion evaluates each query relative to nearby benign patterns rather than imposing a single global support boundary.

A central decision policy combines the branches. Any query assigned to an attack class by Branch 1 is blocked and logged. If Branch 1 predicts `normal` but Branch 2 flags the query as anomalous, the query is held for a conservative response or review. Queries judged normal by both branches are allowed. Table II summarizes the policy.

IV. DATASET AND EXPERIMENTAL SETUP

A. Data Sources

Four public datasets are combined: SQLiV3 (D1), CSIC 2010 (D2), Superviz25-SQL (D3), and SR-BH 2020 (D4). D3 contains approximately 3.69 million rows generated from real `sqlmap` campaigns against a seeded relational schema and includes attack-technique metadata [15]. The D4 subset used in this study contains 527,813 honeypot rows, including 250,285 rows originally tagged as SQL injection. D4's coarse multi-attack labels are re-tagged into SQLi sub-types by rule-based matching, whereas D3's `attack_technique` field is retained because each attack campaign is generated for a known technique. D2 and D4 originate from HTTP traffic; query-string/body parameter content is extracted and treated as query-level input rather than as native database-proxy SQL traces.

TABLE III
PUBLIC DATA SOURCES AND THEIR ROLES IN THE EVALUATION.

ID	Source	Primary role
D1	SQLiV3	SQL-query diversity and benign/attack examples
D2	CSIC 2010	HTTP-derived benign enrichment and anomaly evaluation
D3	Superviz25-SQL	≈3.69M `sqlmap`/benign rows; attack campaigns and complex SQL
D4	SR-BH 2020	527,813-row study subset with SQLi-enriched honeypot traffic

D3 contributes both attack diversity and complex benign SQL, including joins, sub-queries, multi-table analytical statements, and geographic functions. A source-removal diagnostic confirms that this heterogeneity is material: excluding a benign source from Branch 2 training raises FPR on that source's own held-out traffic to 64.8–79.8%, whereas the full four-source model remains in the 2.8–7.2% range across sources. Thus all four benign sources are retained.

B. Branch 1 and Branch 2 Datasets

For Branch 1, content-based filtering removes mislabeled attacks from the nominally normal pool, and canonical query text is deduplicated before splitting. D3 attack rows carry an `attack_id` identifying the generating `sqlmap` campaign. Because rows within one campaign can differ mainly in a narrowing comparison bound, D3 is split by complete campaign rather than by individual rows. D1, D2, and D4 use stratified row-level splits because equivalent campaign identifiers are unavailable. The final Branch 1 benchmark contains 90,000 rows over six classes, undersampled to 15,000 per class, with 71,705 training and 18,295 test instances.

D3 campaigns also separate reconnaissance and exploitation stages. In the source experiments, Branch 1 detects exploitation-stage queries at 100.0% and reconnaissance-stage queries at 94.0%. This difference is relevant because reconnaissance traffic is lexically closer to legitimate database probing, and a campaign-level split can place a disproportionate number of recon-heavy campaigns in one held-out class partition.

Branch 2 pools clean benign rows from all four sources. After source-specific filtering, deduplication, and HTTP-parameter extraction where applicable, the benign pool contains 72,162 rows (57,321 train and 14,841 test). A separate held-out set of 370,002 confirmed-SQLi rows spanning the four sources is used only for anomaly evaluation.

C. Evaluation Protocol

Branch 1 is evaluated with macro-F1 and per-class precision, recall, and F1. Branch 2 is evaluated with FPR, detection rate (DR), AUC, and a threshold sweep. All experiments use seed 42, and latency is measured on a single RTX 3050 (6 GB) workstation.

Unseen-family generalization is evaluated for four attack classes: union-based, error-based, boolean-blind, and time-blind SQLi. For each target class c, all labeled examples of c

TABLE IV
BRANCH 1 ARCHITECTURE COMPARISON.

Model	Macro-F1	p50 latency
TF-IDF + Logistic Regression	0.898	0.5 ms
TF-IDF + LightGBM	0.962	3.0 ms
DistilBERT	**0.965**	2.8 ms*
CNN + SQL tokenizer	0.749	**0.3 ms**

*GPU.

are removed from supervised training, Branch 1 is retrained on the remaining labels, and the withheld class is evaluated as unseen. Branch 2 is unchanged across folds because it is trained only on benign traffic. Both branches therefore score the same withheld queries in each evaluated fold.

For each unseen class c, we report

$$\text{Miss}_{B1}(c) = \frac{|\{q \in c : \hat{y}_{B1}(q) = \text{normal}\}|}{|c|}, \quad (1)$$

$$\text{DR}_{B2}(c) = \frac{|\{q \in c : B2(q) \text{ is anomalous}\}|}{|c|}, \quad (2)$$

$$\text{Coverage}(c) = \frac{|\{q \in c : B1 \text{ or } B2 \text{ flags } q\}|}{|c|}. \quad (3)$$

These measures separate two aspects of generalization: whether the supervised model preserves an attack/non-attack boundary for a withheld family, and whether benign-only anomaly scoring recovers queries that fall into the supervised normal region.

V. EXPERIMENTAL RESULTS

A. Supervised Multi-Class Classification

Table IV compares the four supervised candidates. DistilBERT reaches the highest macro-F1 (0.965), narrowly ahead of LightGBM (0.962). Logistic Regression is substantially smaller and faster, but trails DistilBERT by 6.7 macro-F1 points. The independently retrained final DistilBERT model reaches 0.9699 macro-F1 on the 18,295-query held-out test set.

Table V gives the final DistilBERT class-wise results. Union-based, error-based, and boolean-blind SQLi exceed 0.99 F1, while `normal`, `time_blind`, and `stacked` remain slightly harder. The two attack classes that benefit most from DistilBERT relative to Logistic Regression are `stacked` and `time_blind`: Logistic Regression obtains 0.700/0.812 F1, whereas DistilBERT obtains 0.932/0.973. DistilBERT's 2.8 ms inference latency remains compatible with per-query proxy processing on the measured GPU, although the 256 MB model and roughly 1,970 s training time are substantially larger than the linear baseline.

The lower `stacked` score is consistent with the campaign-aware split. Its test partition contains only 8 of D3's 45 stacked campaigns and is relatively recon-heavy, concentrating the harder, less payload-explicit queries in the held-out set. Per-class AUC remains at least 0.964 across all six classes, indicating that the remaining closed-set errors are concentrated mainly in class boundaries rather than a broad collapse between benign and malicious traffic.

TABLE V
FINAL DISTILBERT PER-CLASS RESULTS ($n = 18{,}295$).

Class	Prec.	Recall	F1	Support
normal	0.890	0.973	0.930	3,000
union-based	1.000	0.994	0.997	3,027
error-based	0.999	0.989	0.994	3,033
boolean-blind	0.990	0.997	0.994	3,016
time-blind	0.984	0.962	0.973	3,007
stacked	0.959	0.905	0.932	3,212

TABLE VI
BRANCH 2 ANOMALY DETECTOR COMPARISON.

Detector	FPR	DR	AUC
One-Class SVM	**0.20%**	11.08%	0.579
Isolation Forest	1.22%	47.23%	0.903
LOF	5.27%	**78.42%**	**0.915**

B. Benign-Only Anomaly Detection

Table VI compares the three anomaly algorithms. One-Class SVM reaches a very low FPR but detects only 11.08% of attacks. Isolation Forest improves DR to 47.23% at 1.22% FPR. LOF reaches the highest AUC (0.915) and DR (78.42%) at a 5.27% FPR, corresponding to 782 false positives among 14,841 benign test queries.

A 21-point threshold sweep shows that FPR falls sharply over the first operating points while DR decreases more gradually; beyond roughly the middle of the sweep, additional FPR reduction flattens while attack detection continues to fall. The reported LOF operating point (contamination $= 0.05$) lies in the steeper part of this trade-off. LOF's higher detection also increases runtime because scoring requires a nearest-neighbor lookup against the retained training feature matrix. Branch 2 costs approximately 15–18 ms/query on the reported workstation. Together with Branch 1, the component-level budget is approximately 18–21 ms/query.

C. Unseen-Family Generalization

Table VII reports the leave-one-attack-family-out experiment. For Branch 1, the relevant event is whether a withheld-family query is mapped to `normal`; assignment to any learned attack class still produces an attack flag. Branch 2 is evaluated independently on the same withheld queries, and combined coverage is the union of the two alarms.

Union-based, error-based, and time-blind SQLi retain strong transfer under class exclusion. Branch 1 misses at most 2.6% of these queries, while Branch 2 independently detects 98.3–99.1%. Combined coverage reaches 99.8–100%. For these structurally distinctive payloads, the supervised branch largely preserves an attack/non-attack boundary even without the withheld class label; most queries transfer to another learned attack category rather than to `normal`.

Boolean-blind SQLi behaves differently. When the family is absent from supervised training, Branch 1 maps 86.5% of its

TABLE VII
LEAVE-ONE-ATTACK-FAMILY-OUT RESULTS FOR THE FOUR EVALUATED SQLI FAMILIES.

Unseen attack	B1 miss ↓	B2 DR ↑	Combined ↑
union-based	2.6%	99.1%	99.8%
error-based	0.0%	98.3%	**100.0%**
boolean-blind	**86.5%**	87.8%	88.8%
time-blind	0.1%	99.0%	**100.0%**

queries to `normal`. Many boolean-blind probes differ from benign conditional queries by small predicate or comparison changes, so their malicious intent is less explicit in an isolated statement. Branch 2 independently detects 87.8% of the withheld boolean-blind queries, demonstrating that its local structural features contribute information not captured by the supervised decision boundary. The union of both branches reaches 88.8%, a large increase over Branch 1 alone, but 11.2% of boolean-blind queries remain undetected. This is the largest residual among the four evaluated withheld families.

The contrast with the standard test split is particularly informative. DistilBERT reaches 0.994 F1 on `boolean_blind` when the class is represented in training, yet the leave-one-attack-family-out experiment exhibits a large miss rate when that family is withheld. Thus closed-set class recognition and unseen-family transfer measure different capabilities: the former evaluates discrimination after exposure to the class, whereas the latter tests whether the learned attack representation extends beyond the observed label set.

D. Complementarity of the Two Branches

The unseen-family results show that complementarity is attack-dependent. For distinctive unseen families, Branch 1 already flags almost every query, so Branch 2 mainly provides an independent confirmation signal. For boolean-blind SQLi, the situation reverses: the supervised branch frequently falls into the normal region, and anomaly scoring supplies most of the additional coverage. The remaining 11.2% gap identifies a shared failure region in which a single query is sufficiently close to benign traffic for both models.

This overlap is more informative than aggregate ensemble accuracy alone because both branches are measured on identical unseen instances. It separates three cases—transfer by supervised classification, recovery by benign-only anomaly detection, and joint failure—and makes the generalization behavior of each evaluated SQLi family directly observable.

VI. DISCUSSION AND LIMITATIONS

The results indicate that supervised classification and benign-only anomaly scoring capture different aspects of SQLi structure. Branch 1 provides accurate attack typing for categories represented in training, whereas Branch 2 does not depend on attack labels and can contribute when the supervised label space is incomplete. This distinction is most visible for boolean-blind SQLi: a class that reaches 0.994 F1 in the closed-set setting is frequently mapped to `normal` when withheld from supervised training, while the benign-only branch recovers a large fraction of those queries.

LOF's 5.27% FPR defines an important operational trade-off. A Branch 2-only alarm is therefore better interpreted as an anomaly signal for logging, rate limiting, or review rather than as an attack-subtype prediction. The operating threshold can be adjusted along the measured FPR–DR curve according to the relative cost of false positives and missed attacks. The present experiments report branch-specific benign performance and attack-side combined coverage; deployment-level calibration of the joint decision policy remains traffic dependent.

Dataset construction introduces several constraints. D3 uses campaign-level splitting to reduce near-duplicate leakage, but equivalent campaign identifiers are unavailable for D1, D2, and D4. The combined benchmark therefore improves diversity without constituting a fully environment-held-out evaluation. In addition, content filtering found mislabeled attacks in the nominally normal data, and a small 30-sample audit estimated approximately 13% label noise in the `boolean_blind` catch-all class. These factors may contribute to the remaining error rates.

The threat model is query-level and targets SQLi evidence available near the application–database boundary. Because D2 and D4 contribute HTTP-derived parameter content rather than native proxy traces, the experiments evaluate the detection logic across heterogeneous query representations rather than a complete live-proxy implementation. Second-order and out-of-band SQLi and non-SQL web attacks are outside the current scope, and deliberate adversarial transformations such as WAF-A-MoLE [17] are not included. The residual on unseen boolean-blind queries also points to information that is difficult to recover from isolated statements alone: repeated probes, changing comparison bounds, and timing behavior become more informative when correlated across a session. Session-level modeling, independently captured database-proxy traffic, and live throughput evaluation are natural extensions of the present query-level study.

VII. CONCLUSION

This paper presented a two-branch query-level SQLi detector that combines supervised six-class classification with benign-only anomaly detection for deployment near the application–database boundary. The selected DistilBERT model reaches 0.9699 macro-F1 on the standard held-out test split, while LOF reaches an AUC of 0.915 and detects 78.42% of anomalous queries at 5.27% FPR. Leave-one-attack-family-out evaluation over four SQLi families reveals strongly attack-dependent generalization. Union-based, error-based, and time-blind SQLi retain 99.8–100% combined coverage, whereas withholding boolean-blind SQLi causes an 86.5% Branch 1 miss rate. Benign-only anomaly scoring raises coverage of this family to 88.8%, with an 11.2% residual gap. The results show that near-perfect closed-set classification does not guarantee transfer to an unseen attack family and that benign-only structural anomaly detection can provide complementary coverage when supervised transfer fails. Future work will

extend the study to session-aware correlation, independently captured proxy traffic, adversarial transformations, and live throughput measurements.

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